

- <https://www.techinterviewhandbook.org/>

1. Why do you want to work for X company?
2. What are you looking for in your next role?
3. Tell me about a time when you had a conflict with a co-worker.
4. What project are you currently working on?
5. What is the most challenging aspect of your current project?
6. How do you tackle challenges? Name a difficult challenge you faced while working on a project, how you overcame it, and what you learned.
7. What excites you about your work?
8. What frustrates you?
9. Imagine it is your first day here at the company. What do you want to work on? What features would you improve on?
10. What are the most interesting projects you have worked on, and how might they be relevant to this company's environment?
11. Tell me about a time you disagreed with your manager.
12. Talk about a project you are most passionate about, or one where you did your best work.
13. What does your best day of work look like?
14. What is something that you had to push for in your previous projects?
15. What is the most constructive feedback you have received in your career?
16. Tell me about a time you met a tight deadline.
17. Time management has become a necessary factor in productivity. Give an example of a time-management skill you've learned and applied at work.
18. What strengths do you think are most important for your job position?
19. What words would your colleagues use to describe you?
20. What would you hope to achieve in the first six months after being hired?
21. What does "belong anywhere" mean to you?
22. What large problems in the world would you solve today?
23. If you had an unlimited budget and you could go somewhere, where would you go?
24. Share one of your trips with us.
25. What is the most challenging project in or out of school that you have worked on in the last 6 months?
26. What's something you'd like to remove from the Airbnb experience?
27. What is something new that you can teach your interviewer in a few minutes?
28. Tell me about why you want to work here.
29. What is the best gift you have ever given or received?
30. Tell me about a time you were uncomfortable and how you dealt with it.
31. Explain a project that you worked on recently.
32. Tell me something about yourself and why you'd be a good fit for the position.
33. Name a situation where you were impressed by a company's customer service.
34. How did you work with senior management on large projects as well as multiple internal teams?
35. Tell me about a time you had to give someone terrible news.
36. If you were a gerbil, which gerbil would you be?

37. What excites you about the company?
38. How does Airbnb impact our guests and hosts?
39. What part of our mission resonates the most with you?
40. How do you deal with a failed deadline?
41. Why do you want to work at Amazon?
42. Tell me about a situation where you had a conflict with a teammate.
43. In your professional experience, have you worked on something without getting approval from your manager?
44. Tell me a situation where you would have done something differently from what you actually did.
45. What is the most exceedingly bad misstep you've made at any point?
46. Describe what Human Resources means to you.
47. How would you improve Amazon's website?
48. Take me through a product you launched from start to end.
49. What's the biggest achievement in your previous projects?
50. Tell me about a recent failure and what you learned from the experience.
51. Why do you want to work at ByteDance?
52. What makes you a good fit for this position?
53. Talk about your favorite project.
54. If you were hired here, what would you do?
55. State an experience about how you solved a technical problem. Be specific about the diagnosis and process.
56. What is your biggest strength and area of growth?
57. What are your salary expectations?
58. Tell me about a time your work responsibilities got a little overwhelming. What did you do?
59. Give me an example of a time when you had a difference of opinion with a team member. How did you handle that?
60. Tell me about a challenge you faced recently in your role. How did you tackle it? What was the outcome?
61. Tell me about your most interesting/challenging project to date.
62. Why Lyft? What are you looking for in the next role?
63. What is something 90% of people disagree with you about?
64. What is broken around you?
65. How do you deal with difficult coworkers? Think about specific instances where you resolved conflicts.
66. How did you win over the difficult employees?
67. Tell me about an analytical problem that you have worked on in the past.
68. If you were in charge of picking projects for Palantir, what problem would you try to solve?
69. What are some of the best and worst things about your current company?
70. What would your manager say about you?
71. Describe Palantir to your grandmother.
72. Teach me something you've learned.

73. Tell me a time when you predicted something.
74. If your supervisors were to rate you on a scale of 1-10, what would they rate you?
75. What was the most fun thing you did recently?
76. Tell me the story of how you became who you are today and what made you apply to Palantir.
77. Tell me something about your internship.
78. Why do you want to join Slack?
79. Tell me about your past projects.
80. Explain to me your toughest project and the working architecture.
81. Apart from technical knowledge, what did you learn during your internship?
82. If someone has a different viewpoint on a project, such as using a different programming language, how would you handle this situation?
83. What are your most interesting subjects and why?
84. Did you find any bugs in Slack?
85. What is your favorite feature and why?
86. What have you built?
87. What is the hardest technical problem you have run into?
88. How did you solve it?
89. How do you stay up to date with the latest technologies?
90. Explain a project that you worked on recently that was difficult.

- **<https://www.coursera.org/gb/articles/software-engineer-interview-questions>**

1. What programming languages are you familiar with?
2. What is the software development process?
3. Describe the Agile methodology and its benefits.
4. What are the characteristics of these data structures?
5. What are the differences between SQL and NoSQL databases?
6. Why do you want to work here?
7. Describe a problem you've encountered in software development and how you solved it.
8. How do you handle conflicts or disagreements within a team?
9. What would you do if you were assigned a project with a technology you're not familiar with?
10. How do you stay updated on the latest industry trends and technologies?

- **<https://www.intuit.com/blog/innovative-thinking/60-engineering-interview-questions-to-expect-and-12-to-ask-during-your-job-search/>**

Role-Specific and Background Questions:

1. Why did you apply for this particular role?
2. How did you hear about this position?
3. Are you applying for other jobs?
4. Why do you think you're a good fit for our company?
5. What do you think our company can offer you?
6. Which words would you use to describe yourself?

I would describe myself as a dedicated, problem-solving, and curious individual. I am dedicated because when I commit to a project or task, I ensure it gets done to the best of my ability. I am a problem-solver, as I enjoy identifying solutions and overcoming obstacles in my work. Finally, I'm curious, meaning I am always eager to learn new technologies and methodologies, and constantly strive to improve my skills and knowledge in the field of software engineering.

7. What are your biggest strengths? What are your flaws?

One of my biggest strength is that I'm good at problem-solving. I always break down complex problems into smaller, manageable tasks and finding efficient solutions. This is really useful in debugging and troubleshooting code, and finding innovative solutions for our software needs.

Another strength is my adaptability. In the fast-paced tech industry, things change quickly, so being able to adapt to new technologies, methodologies or project requirements is crucial.

As for flaws, I can be a bit too straight forward at times and sometimes that hurts people. But I'm trying to stop myself from being overly direct.

8. What do you like most about engineering? What do you dislike?

What I like most about engineering is the process of problem-solving. I enjoy the challenge of taking a complex issue, breaking it down into manageable parts, and then coming up with an effective solution. It's like a puzzle to me, and the satisfaction I get when I find the solution is immensely gratifying.

As for dislikes, there can be times when the process of debugging can be quite tedious, especially if the issue is not apparent and requires a significant amount of time to resolve. However, I view it as part of the journey and a learning experience, so I try to maintain a positive perspective and approach each challenge as a chance to learn and grow.

9. What motivates you in your day-to-day work?

What really motivates me is the opportunity to learn and grow. The tech world is continuously evolving with new tools, technologies, and best practices, and I find it exciting to keep pace with these changes. Also in my next role, I'm looking for that place that could provide me lots of exciting and interesting challenges at the same time with growing my worth to the company.

10. What is your ideal work environment?

My ideal work environment is one that revolves around the team where the focus is on learning, working and growing together to take a team and the company to new heights. I can say I'm a great team player so I'm always keen on teamwork and keeping a good relationship with my colleagues but that doesn't mean that I cannot work independently. While I like team collaboration I can also work well independently. So no specific requirements for me based on the ideal work environment.

11. Can you tell us more about your work ethic?

Honestly, my work ethic is pretty steady and consistent, I like to take full ownership of what I'm working on, so once I pick up a ticket, I make sure I follow it through end to end. I usually start by understanding the problem clearly, asking questions early so I don't go in the wrong direction. Then I plan a bit, break things down, and just start building step by step.

I don't really wait around or push things last minute. I try to stay ahead of deadlines, and if something is blocked, I'll communicate early and try to solve it quickly.

I also care about quality. Like, I don't just want it to work, I want it to be clean and easy for the next person to understand too.

And yeah, I'm pretty collaborative too. I don't hesitate to jump on a quick call or review someone's code if it helps the team move faster.

12. Where do you see yourself 10 years from now?

13. What are your current goals?

My current goals are primarily focused on professional growth and learning. At first I aim to grow as an expert programmer and then after that I want to become someone who manages teams and provides technical and professional solutions to the team.

14. Why did you pursue a degree in software/computer/data engineering?

15. What are your salary expectations?

Previous Experience and Background:

16. What is your educational background?

17. Are you looking to obtain a higher educational degree (e.g., Ph.D.)?

18. Which coding languages do you know, and which ones have you used?

19. Did you complete any courses or obtain technical certifications in the last year?

20. Did you learn or improve any engineering skills in the last six months?

21. How do you keep updated with the latest technology?

22. How familiar are you with artificial intelligence (AI) and large language models (LLMs)?

23. What software packages/tools/suites are you familiar with?

24. What were your most recent experiences in engineering?

25. What were your tasks and responsibilities in your previous role?

26. In which industries have you worked during your career?

27. What was your biggest accomplishment during your career?

28. Can you provide an example of how you have applied your technical knowledge practically?

29. On your previous projects, what problems or weaknesses were you able to identify and solve?

30. How did you work with a team to meet a shared goal?

Technical Questions:

31. Describe your process when writing a piece of code, including the tools you use and how you meet deadlines.

32. Can you explain why wireless signals lose power over distance and how to address this issue in a large, multi-storied office building?

33. What is the significance of RESTful APIs in web development?

34. Can you describe a time you optimized an inefficient piece of code?

35. What are the principles of agile software development?

36. Can you summarize the differences between SQL and NoSQL databases?
37. What steps do you take to ensure the security of your code?
38. Can you describe the three most important steps to optimizing a website for mobile?
39. Are you familiar with service-oriented architecture (SOA)?
40. Can you tell me the differences and practical applications of vertical scaling and horizontal scaling?
41. What is the difference between synchronous and asynchronous programming?
42. How do you implement error handling in a large-scale distributed system?
43. Which programming languages would you use to analyze a large dataset and why?
44. What experience do you have with cloud computing platforms like AWS or Azure?
45. What are the advantages and disadvantages of using Kubernetes versus Docker containers?

Soft Skills Questions:

46. How do you ensure effective collaboration within a diverse team?
47. Have you ever mentored or coached less experienced team members?
48. If you didn't choose an engineering career, what would you be doing today?
49. How do you ensure quality in your deliverables on a day-to-day basis?
50. How would you handle a difficult client who is particularly disappointed with the engineering project you're working on?
51. A few days before an important deadline, you realize you won't be able to meet it in time. What do you do?
52. A project you're working on is facing a serious issue. Your employer blames you, but you know it's the fault of an inexperienced junior colleague. How do you handle the situation?
53. What is the most negative feedback you've received from a supervisor? And how did you react?
54. What personal characteristics do you feel are the most important to be successful in your line of work?
55. Have you ever led a project? Why were you chosen for this role?
56. What processes or strategies do you employ to minimize the risk of making mistakes while you work?
57. How would you describe a perfect team?
58. What's the biggest challenge you might face during your job?
59. What characteristics do you expect from your ideal manager?
60. Given the choice, would you prefer to work on a small team, a large team, or alone?
61. Questions You Should Ask the Interviewer:
62. Can you describe the team I'll be working with?
63. Can you outline a typical day in this role?
64. What projects will I be working on in the immediate future?
65. What tools, software, platforms, or suites does the team use?
66. What metrics are used to measure success in the company and within the engineering team?
67. How do managers or supervisors handle performance reviews and feedback?
68. How is communication managed between management and staff (emails, status reports, weekly meetings)?

69. What are the current biggest challenges facing the engineering team, and how can I contribute to solving them?
70. How does my role contribute to the company's overall success?
71. Does the company offer opportunities for training or professional development? If so, what are they?
72. How does the company support work-life balance?
73. What are the next steps in the interview process?

- <https://dev.to/fpaghar/25-hr-interview-questions-for-developer-with-sample-12c9>

1. Tell me about yourself.

I have been experiencing in programming for around 10 years. The last company I was working at is ... company. In my most recent front-end developer position, I'm responsible for designing responsive web pages mainly using HTML, CSS, Sass, JavaScript, ES6 Syntax, ReactJS, Redux, React hooks, Next.js, TypeScript, and styled components to support performance perfectly. My role in ... company includes: Collaborated with innovation, BE, CMS, and QA members to implement ... and ... products.

Performs complex bug verification, release testing, and data support for assigned products.

Research problems discovered by QA or product support and develop solutions.

Helping to define and shape the architecture and frameworks used in the front-end application, or updating packages or frameworks.

2. What are your biggest strengths, abilities, features, and characteristics?

My main ability in this work is effectively handling projects and deadlines. I've been known as a passionate, responsive, and committed person who provides a suitable solution with high performance under strict conditions. I am keen on teamwork and keeping a good relationship with my colleagues, but I can also work well independently. Listening to criticisms and feedback, especially if they are constructive, is one of my strengths.

3. What do your friends and colleagues think about you?

It is something that I've never thought about, but I think they would say I am a hardworking and committed person. They would say I am a positive and supportive person. I can say I have 2 different characters: with my colleagues, I am a serious person, and they say I have a strong relationship, and I am good at delivering projects. But among my friends, I've been known as an honest and fun person.

4. Why did you choose our company for cooperation?

As I visited your website, I found out that you are working in healthcare industry, which is really interesting to me. I've found that you are looking for a full-stack developer who is skilled in both frontend and backend part. So considering my skills and abilities, I believe this position is suitable for me. Working in a start-up company seemed fantastic, and I am excited about the challenges. Firstly, this job is an ideal match for my skills and experience. Secondly, I know ... the company is one of the fastest-growing companies in the world, and I want to be part of that.

5. Why did you leave your last job?

To be honest, I loved my previous workplace because I learned a lot from some amazing people. But I started thinking that it's time to expand my horizon and take the next step of my career. So I started looking for a new place that could provide me with lots of exciting and interesting challenges and I think your company will be the perfect place for me to utilize myself as a professional and personal growth.

6. What are your career goals/prospects?

In the first 2-3 years, I see myself as an expert programmer who can have a good effect on the team and company. For the 2-3 years after that, I see myself as someone who can manage the team and offer technical and professional solutions for the company.

7. What is your greatest weakness?

I certainly admit that I am not perfect. When I was a leader in the team, I was impatient with the irresponsibility of some colleagues. For example, someone who could finish his task but didn't get it done because he had time to wait, and part of the project was waiting for him to finish. I think it is annoying. But I try not to let it affect my performance. Well, as I mentioned before, I do tend to get frustrated if I don't see progress in my work.

8. Are you applying for other jobs?

Of course, I am. I am looking for the best place to put my skills into practice, and so far, this place looks excellent.

9. What do you know about our organization?

I know that your company specializes in [specific industry or product] and has been recognized for its innovation in [specific area]. I have researched your recent achievements, including [specific project or success], and I am excited about the opportunity to contribute to your team's growth and success.

10. What's your dream job?

My ideal job position is one where I have more responsibilities and room for growth. I am looking for a role that allows me to develop my qualifications further and be successful in my job, both for my satisfaction and for the company's success.

11. Why should we hire you? What makes you a good fit for this role?

I have good experience and passion for this situation. I also learned through my research that your company values X, Y, and Z, and I know without a doubt my experience in this area will sustain company standards and lead the team to greater success. I read about this position on your website, and I think that I can be a suitable candidate for this position.

12. How long would you expect to work for us if hired?

I'd like to be here for a long time, as long as you permit me, and I believe that if I do my tasks properly, I will remain in your company. I worked for 5 years in my previous company. That demonstrates my loyalty. So, as long as the job is challenging and I have growth and training potential, I will be happy to work here. I immigrated, and I am looking for a new workplace where I can stay for a long time.

13. Are you willing to work overtime, nights, or weekends?

I had this experience 2 years ago when we got the first project of a government client. We didn't have enough time, and I worked overtime at night, not going home for a week. After that, I lost some motivation for working overtime because even though I kept the

project alive, I just received some false promises. I love my job, and I believe that when a project deadline is coming up, everyone must pull together to finish it.

14. Do you consider yourself successful?

Yes, because everywhere I worked, my employer and colleagues were very satisfied with me, because I've done my best to be effective.

15. Explain how you would be an asset to this organization.

The capabilities I have in my field can be very useful for the improvement of quality in my department, giving the overall benefit to the company. I've always been a trustworthy person, and that's one of the important reasons.

16. What is more important to you: the money or the work?

I believe they are equally important to our well-being. But I am not looking to get rich, and I am only looking for an opportunity where I can prove my worth to the company, and this is my priority. I think work follows money, and money follows work. However, I can adjust for money to some extent if the work is challenging and provides job satisfaction.

17. Tell me about a problem you had with a supervisor.

My supervisor was not an on-time person, and many times this caused problems in the project. This was the only thing I had problems with my supervisor.

18. Tell me about your ability to work under pressure.

It's funny to say I enjoy this situation sometimes, but not always. Pressure is a part of my work. When there is an imperative deadline, I put my focus on my work, which has helped me to show my special ability. I stay focused and work fast under pressure.

19. What motivates you to do your best on the job?

I want to be successful in my job, both for my satisfaction and for my employer. When I see that my employer is satisfied, I will be happy, and I like working with someone who lets me do my job to the best of my knowledge and abilities.

20. How much is your salary expectation?

Thanks for asking. I am not looking to get rich, and I am only looking for an opportunity where I can prove my worth to the company, and this is my priority. I am looking for a position that pays between 60,000 and 70,000 annually, but I am open to negotiating depending on my responsibilities and benefits.

21. If you have 2 equal positions in salary and benefits, which one would you opt for?

My choice will depend on the company grade, the number of development teams, and the friendly relationships between company members. Also, in my opinion, the standard of assignments and reviewing responsibilities is important. I mean, how we apply Scrum, technologies, and design patterns used in the company, and company investments for using new technologies are very important to me.

22. What was your last responsibility?

I worked as a backend and frontend developer for 5 years. I was in charge of monitoring the members of the team and solving technical problems. I also participated in analysis, particularly in financial analyses, and made some effective solutions. In the latest projects, we used .NET Core and Angular with the DDD pattern.

23. Would you like to travel?

Yes, of course. Traveling, especially in nature, relieves me of stress and tiredness from work. I am keen on traveling in a group.

24. If you gain this job position, do you have any problem with commuting?

No, I am going to move to my new place, and there isn't any problem.

25. Have you ever failed a project?

Yes, it was an online taxi project, and unfortunately, despite it being interesting to me, the project failed. There were many reasons for failing the project, and we cannot point to one person. One of the main reasons was our product owner. He could not manage his time and relationship with the client to gain accurate information, which ultimately led to the client's dissatisfaction. The project was put aside by the client. Additionally, international companies began their online taxi projects in Iran, which was another factor.

26. Can you explain a little about the software product you worked on?

I think the provincial government project was a challenging one. The main challenge was finding the most effective way to update the client's data and database structure. The solution was to store an updated copy of data on clients' systems, thus reducing the server load, and the clients had distinct databases. We used a script code to update data and notify clients when updates were completed.

27. Tell me about your experience working with women in the workplace.

I think it is a commonplace issue, and in my last position, I had high interaction with my colleagues, both men and women.

- <https://igotanooffer.com/blogs/tech/software-engineer-behavioral-interview-questions>

1. Tell me about a time you worked well within a team

Situation: "In my previous role, our team had to develop a real-time data analytics platform for a client with a tight deadline."

Problem: "We had to integrate multiple complex systems, including data ingestion, processing, and visualization."

Solution: "I facilitated daily stand-up meetings as the Scrum Master, ensuring everyone knew their tasks. We had a brainstorming session to tackle integration issues."

Impact: "The platform was delivered on time and exceeded the client's expectations."

Lessons: "I learned that effective communication and collaboration are key to overcoming technical challenges."

2. Describe a situation where you had to resolve a conflict on a team. How did you solve it?

Situation: "In a fintech startup, two engineers had a clash due to communication issues."

Problem: "One engineer, Rob, was unhappy with Benji's rude behavior."

Solution: "I facilitated a private conversation between the two, where they cleared the air. I encouraged Benji to apologize and improve his communication."

Impact: "The working relationship between Rob and Benji improved significantly."

Lessons: "Conflict resolution often requires empathy and open communication to understand both sides."

3. Tell me about a time you failed at work

Situation: "While working on a booking feature, I rushed and overlooked testing, which led to a bug."

Problem: "The bug caused issues on the client's website, affecting user experience."

Solution: "I sought help from my manager and a senior colleague. We informed the client and worked overtime to fix the issue."

Impact: "Though we missed the deadline, the client appreciated our transparency and effort."

Lessons: "I learned never to skip thorough testing and to always communicate openly with clients about issues."

4. Why do you want to work here?

Answer: "I am excited about the mission and values of your company, especially the focus on [insert company's value here]. I want to contribute my skills in [specific skill/technology] and work on [specific project or company initiative] because I see it as a great match for my background and career goals."

5. Tell me about a time you showed leadership

Situation: "At my previous job, I mentored a new intern who struggled with some of the tools we used."

Problem: "The intern wasn't able to complete tasks and felt isolated."

Solution: "I decided to work closely with the intern, helping him learn the tools and giving him smaller tasks to build confidence."

Impact: "The intern became a valuable team member and was hired full-time after his internship ended."

Lessons: "I learned that leadership involves supporting others to grow, not just taking charge of tasks."

6. Tell me about an interesting project you've worked on recently

As part of your preparation, you should look back at some successful projects and pinpoint the actions you took that demonstrate certain skills that the job description is looking for. If possible, give some metrics to quantify your impact.

Not everything about the project has to be a total success - talk about what you learned and how you'll be able to do better next time.

Example answer:

(Situation)

At my previous company, I had the opportunity to lead a crucial project with guidance from my manager.

(Problem)

The challenge was to create a seamless and efficient real-time editor that allowed multiple users to edit a document simultaneously, ensuring synchronization in real-time.

(Solution)

We opted for a microservices architecture, using technologies like Node.js and WebSockets to establish real-time communication. We also employed Operational Transformation algorithms to handle concurrent editing and conflict resolution. A thorough load test was conducted to ensure the system could handle a large number of concurrent users seamlessly.

(Impact)

The project was a success, and the real-time collaborative editor was integrated into the platform, significantly enhancing the remote collaboration experience for our users. We observed a 35% increase in user engagement with the new editor, and user feedback was overwhelmingly positive, with users commenting on its efficiency and ease of use.

(Lessons)

On a technical level, this project reinforced the importance of a scalable and robust architecture. It taught me the significance of comprehensive testing, especially when dealing with real-time applications.

On a personal level, this project gave me a lot of confidence in my ability to lead. Before this, I'd had little experience of leading meetings and taking responsibility for coordinating projects, but since I've led quite a few more. I've got much better at anticipating other people's doubts and seeing where bottlenecks might occur, and I've lost the shyness that used to prevent me from taking an active role in team meetings."

If you can pick a project that you led, like the one above, that's ideal, as it makes it easier to demonstrate leadership skills. If you can't, don't worry, just be sure to demonstrate other soft skills such as collaboration, time management, adaptability, or problem-solving, as well as technical skills.

7. Tell me about a time you faced a really hard problem / a challenge at work

Employers at competitive companies want to know how you'll react under difficult circumstances. Your SWE resume may be full of impressive achievements with quantifiable impacts, but that doesn't necessarily show that you can thrive when the going gets tough.

Don't make the mistake of spending too long explaining the problem. Succinctly outline the situation and the problem/challenge, then spend the bulk of your answer talking about what you did, the impact your actions had, and what you learned.

Example answer:

(Situation)

"In a previous role, I faced a significant challenge while working on a large-scale e-commerce platform.

(Problem)

The system started experiencing intermittent performance degradation during high-traffic periods, severely impacting user experience and potentially resulting in revenue loss.

The problem had two main aspects: optimizing the system to handle a massive influx of users during peak times and identifying specific bottlenecks causing the performance degradation.

(Solution)

In coordination with my manager, I led a comprehensive analysis, utilizing various profiling and monitoring tools to pinpoint performance bottlenecks. Once we identified the critical areas, we orchestrated a focused effort on code refactoring, optimized database queries, and implemented caching mechanisms to enhance response times. Additionally, I worked with my superior to restructure the system's architecture to incorporate load balancing and auto-scaling strategies, ensuring it could handle higher traffic loads.

(Impact)

The efforts resulted in a substantial performance improvement. The platform demonstrated remarkable stability even during peak periods, showcasing a 50% reduction in response times and zero downtime. Users were notably pleased, and the platform successfully handled subsequent peak traffic events smoothly.

(Lessons)

For me, this experience underscored the importance of proactive performance monitoring and scaling strategies. It's worth investing a bit of time earlier on to protect yourself later from unexpected traffic surges and the like.

I also learned a lot from being in meetings with my manager and heads from other departments, as she briefed them on our progress. She was very effective at giving high-level, simplified information that responded to their concerns, rather than getting lost in the technical details, and I definitely took that on board as something to work on myself."

Notice how the answer sets up the situation and problem very quickly, giving the candidate time to focus on the solution, impact, and what they learned. There is a nice balance here between a focus on their individual actions and demonstrating a collaborative approach.

8. Tell me about a time you showed leadership

Across any industry, leadership is one of the qualities that interviewers are likely to assess you on, even if you're not applying for a leadership role. Why? Because a good leader not only does their job well, but also helps others do their job better. Their positive impact on the company is therefore multiplied.

A junior software engineer, for example, won't know how to lead a team of engineers. But there may be an occasion where they can lead a process, or lead a meeting, and doing this well has an impact on the wider team.

If you're applying for a senior or management role, you may be asked a more specific version of the question, such as "Tell me about a time you led a team through a difficult period". Prepare an example that shows you as someone with some key leadership traits: determined, empathetic, decisive, good motivator, etc.

Don't worry if you're more junior and you've never held a leadership position before.

Your answer can come from a situation in which you weren't in a leadership position but demonstrated leadership nevertheless.

Example answer:

(Situation)

Recently, in my current role, an intern joined our team, called Mo. I had a very high workload at the time, and so was grateful to see an extra pair of hands join the team.

(Problem)

However, Mo lacked experience in some of the key tools we used daily, such as GitHub and Django. He was also very shy. After the first week, he hadn't completed many of the tasks he was given, and people stopped asking for his help. Our manager was so busy that he didn't pay much attention to the situation.

(Solution)

I decided to sit with Mo at lunch and ask him about his experience with coding. I gradually realized that while he'd never worked in a team like ours before, he was a

talented programmer. I told him what I was working on, and he seemed confident he could help.

I delegated some of my tasks to Mo over the next couple of days, making it clear that if he didn't understand something, he should just ask. He completed the tasks very well. By now he came to sit with me every lunchtime and was starting to come out of his shell. I used the time he'd saved me to show him some things on GitHub and Django, and I made sure I told everyone about the great job he'd done for me.

(Impact)

The rest of the team started sending him tasks again, and this time, Mo completed the tasks much more successfully, occasionally asking me for help. Three months later, his internship ended, and he joined the team full-time.

(Lessons)

The experience with Mo made me realize that when somebody joins a team, it's really important that they have someone to go to when they don't understand something or have a problem. It also taught me how crucial it is to take the time to ask and listen, instead of dismissing people after a bad first impression."

This candidate demonstrates various leadership skills with this example: in particular, a lot of initiative and empathy. The interviewer could easily imagine this candidate as a future manager who will do a great job helping his team members reach their potential. See more example answers to this question in 5 ways to answer "Tell me about a time you showed leadership".

9. Tell me about a time you had to meet a tight deadline

This question helps employers assess your ability to handle time-sensitive projects and manage competing priorities.

Demonstrate your proficiency in project management and problem-solving, and your commitment to achieving successful outcomes under pressure.

Use an example of a time when the odds were stacked against you - it will make for a more interesting and compelling answer.

Example answer:

(Situation)

"In the final quarter of last year, our team was tasked with delivering a critical module for a high-profile client, promising to revolutionize their user experience. This project had an unusually tight deadline due to an unexpected product launch by the client, leaving us with only three weeks to deliver what would normally be a two-month project.

(Problem)

The short timeframe was further complicated by unforeseen technical complexities that arose during the initial stages. We were faced with a massive amount of coding, rigorous testing, and integration tasks. The pressure was mounting as we couldn't compromise on the quality and functionality of the module.

(Solution)

To navigate through this challenge, we immediately held a team meeting to re-evaluate our project plan. We swiftly identified non-essential features and functionalities that could be temporarily deferred to subsequent versions. We also employed a parallelized

development approach, allowing multiple team members to work simultaneously on different aspects of the module.

We implemented an Agile methodology, conducting daily stand-up meetings to address roadblocks promptly and ensure everyone was on track. I took on the role of a scrum master, ensuring efficient coordination and smooth progress.

(Impact)

Our collective effort and the adjustments we made were transformative. Despite the initial odds, we managed to deliver a fully functional module that exceeded the client's expectations. The success of this module significantly bolstered our relationship with the client and opened up opportunities for future collaborations.

(Lessons)

This experience reinforced the importance of adaptability and collaboration under extreme time constraints. It taught me the significance of quickly reassessing priorities and being flexible in our approach. It also highlighted the power of effective communication within a team."

10. Tell me about your biggest weakness

A classic interview question at all levels, and one that causes much anxiety for many candidates. It shouldn't. It's not a trick question. We all have weaknesses, and as long as your biggest weakness isn't one that is vital for the job, you should try to answer honestly.

The interviewer is testing to see if you have the self-awareness to know what you're less good at, and enough drive to actually do something about it.

So pick a weakness that is real and is relevant to your role, though not something critical. You can also use this answer to show that you're good at taking on feedback - an important skill and one that your interviewer will be looking out for.

You might find it hard to think of a weakness that you're willing to open up about. To help, here's a list of possible weaknesses that could give you the foundation of a strong answer and would avoid red flags.

Possible weaknesses to mention in a software engineer interview:

Not good at delegating: you struggle to delegate tasks effectively due to a desire to maintain control over the project, potentially leading to missed deadlines.

Limited experience with a specific technology: you acknowledge a need to enhance expertise in a particular programming language or framework.

Over-optimizing code: you tend to spend too much time refining and perfecting code before the core functionality is solidified, potentially affecting deadlines.

Difficulty saying no: You find you agree to take on additional tasks, sometimes resulting in an overloaded schedule and affecting your ability to manage your projects effectively.

Overemphasis on independence: You have had a preference for working independently rather than collaborating within a team.

Nervousness speaking: you find it difficult to present technical ideas or updates clearly and engagingly.

Prioritization challenges: You struggle to effectively prioritize tasks and manage time during busy development cycles.

Occasional lack of documentation: You haven't always consistently documented code and processes during fast-paced development.

These are all weaknesses that can hurt work performance, so you don't want to give the impression that it's something that comes with you as "part of the package". Instead, demonstrate that you're already taking steps to improve at it.

Let's take a look at an example to see how that might look. Notice that this isn't technically a behavioral question in that you don't necessarily need to talk about an experience.

Example answer:

"One area I've identified for improvement is my tendency to over-optimize code during the initial development stages. At times, I find myself spending too much time fine-tuning the code for performance before the core functionality is solidified. There were a couple of occasions where this led me to miss deadlines.

To overcome this, I've adopted a more iterative development approach. I focus on getting the fundamental structure and features in place before delving deeply into optimization. This way, I ensure the code's functionality is robust before investing time in fine-grained performance enhancements. I've also sought feedback from senior colleagues to strike the right balance between optimization and timely project delivery. Making this adjustment has helped me to deliver reliable solutions on time, while still maintaining a high standard of code quality. It's still something I need to work on and improve, but I'm definitely going in the right direction."

11. Describe a time you had to prioritize projects under pressure

Top tech companies are fast-paced environments, and hiring managers want candidates who will get the most important work done on time. This question is testing you on your time-management, decision-making, and adaptability, as well as assessing whether you're someone who takes ownership of projects and accepts responsibility for meeting deadlines.

Example answer:

(Situation)

"I was once given just a weekend to fix a critical bug in a live system. The bug was causing intermittent crashes, affecting the user experience for thousands.

(Problem)

There was a lot of urgency, as every minute the system was down meant potential loss of customers and revenue. Additionally, pinpointing the exact cause of the crash was like finding a needle in a haystack.

(Solution)

Our manager quickly assembled a small team, including me. Each person is assigned to a specific aspect of the system that could be a potential cause. We worked around the clock, running tests, analyzing logs, and discussing findings in real-time.

(Impact)

We identified the root cause late Saturday night, and I took the lead in implementing the fix. The system was stable by early Sunday morning. Despite the tight deadline and immense pressure, we managed to resolve the issue, ensuring uninterrupted service for our users.

(Lessons)

This experience underscored the power of collaboration and focused effort under pressure. It taught me that even in the most demanding situations, a well-coordinated team can achieve remarkable results. Additionally, it emphasized the necessity of keeping a cool head and methodically approaching the problem, especially when time is of the essence."

12. Why do you want to work at [Company]?
13. What is your favorite [Company] product?
14. Why did you start coding / want to become a software engineer?
15. Why are you a good fit for our company?
16. Where do you see yourself in 5 years?
17. Tell me about a recent, favorite, or impactful project you worked on.
18. Tell me about the most challenging project you've worked on.
19. Tell me about your greatest accomplishment / main achievements.
20. Tell me about your biggest failure or something you struggled with.
21. Tell me about a time you adapted well to change.
22. Tell me about a time you exceeded expectations.
23. Tell me about a time you showed leadership or led a team.
24. Describe a situation where you had to resolve a conflict or disagreement on a team.
25. Tell me about a time you disagreed with your manager or superior.
26. Tell me about a time you struggled to work with a colleague.
27. Describe a situation where you collaborated with multiple teams.
28. Tell me about a time you and a cross-functional partner disagreed. How did you resolve it?
29. Tell me about a time you took ownership beyond your responsibilities.
30. Tell me about a time you decided with incomplete information.
31. How would you approach solving a complex problem with uncertainty or unfamiliar technology?
32. Tell me about the biggest lessons you've learned in life.
33. Tell me about your values and the environments where you thrive.
34. How do you give back to the community?
35. Airbnb-specific culture questions
 - What are Airbnb's core values, and how do they apply to you?
 - What does "belong anywhere" mean to you?
 - Tell me about a time you made others feel included.
 - Tell me about a time you were a good host.
 - What would you improve or remove from Airbnb?
25. Netflix culture question
 - What do you think about the Netflix culture memo?
26. OpenAI-specific question
 - What are your thoughts on AI safety and the risks of advanced AI systems?

- <https://www.tealhq.com/career-paths/software-engineer-interview-questions>

1. Tell me about a complex technical problem you solved.

Why interviewers ask this: They want to assess your problem-solving methodology and ability to work through ambiguity.

In my previous role, we had a multi-threaded application experiencing intermittent crashes in production. The issue was difficult to reproduce locally, which made debugging particularly challenging. I started by adding structured logging around the areas where crashes were reported, then analyzed the logs to identify patterns in the timing and sequence of operations.

After narrowing the scope, I discovered a race condition in our shared resource access layer. Two threads were attempting to write to the same data structure simultaneously without proper synchronization. I implemented mutex locks around the critical sections and added comprehensive unit tests to verify thread safety.

The fix eliminated the crashes entirely and actually improved overall application throughput by reducing contention. I documented the root cause and the solution in our team wiki to prevent similar issues in the future.

2. How do you ensure code quality in your work?

Why interviewers ask this: They want to understand your commitment to engineering standards and your familiarity with quality practices.

I take a multi-layered approach to code quality. First, I write unit tests alongside my code — I aim for meaningful test coverage that focuses on business logic and edge cases rather than hitting arbitrary coverage numbers. I use test-driven development when working on complex algorithms or critical paths.

Second, I actively participate in code reviews. I review others' code carefully and welcome thorough reviews of my own work. I've found that code reviews catch not just bugs but also design issues and opportunities for simplification.

Third, I rely on automated tooling — our CI/CD pipeline runs linting, static analysis, and our full test suite on every pull request. I've also introduced SonarQube on a previous team to catch code smells and potential vulnerabilities early. Finally, I maintain clear documentation for complex logic so that future developers can understand the intent behind the code.

3. How do you approach learning a new technology quickly?

Why interviewers ask this: They're evaluating your adaptability and self-directed learning skills — essential traits in a fast-moving field.

When our team decided to migrate from Angular to React, I had about two weeks to become productive. I started with the official React documentation to understand the core mental model — components, state, hooks, and the virtual DOM. I then completed a focused online course that covered practical patterns.

Rather than just reading, I immediately started building. I created a small side project that replicated key features of our existing application in React. I also paired with a colleague

who had React experience, which accelerated my learning significantly. Within the two-week window, I was contributing to production React code and participating meaningfully in architectural discussions about our new frontend.

4. Describe your debugging process for an unfamiliar system.

Why interviewers ask this: They want to see systematic thinking and your ability to work effectively with code you didn't write.

When I encounter a bug in an unfamiliar system, I follow a structured approach. First, I review the architecture documentation and codebase structure to understand the system's overall design and data flow. Then I reproduce the issue in a controlled environment — this is critical because you can't fix what you can't observe.

Once I can reproduce the bug, I use a combination of logging, breakpoints, and tracing to follow the execution path. I narrow the scope systematically — binary search through the call stack, essentially. I check recent commits related to the affected area, review any related test failures, and examine logs for anomalies.

After identifying the root cause, I write a failing test that captures the bug before implementing the fix. This ensures the issue doesn't regress and adds to the system's test coverage.

5. How do you handle technical debt?

Why interviewers ask this: They're assessing your ability to balance short-term delivery with long-term code health.

Technical debt is inevitable in any active codebase, and I think the key is managing it intentionally rather than ignoring it. When I encounter or introduce technical debt, I document it explicitly — usually as a ticket in our backlog with context about why it was incurred and the potential impact.

I advocate for allocating dedicated time each sprint to address technical debt, typically 10–20% of the sprint capacity. I prioritize based on impact: debt that affects developer velocity, system reliability, or security gets addressed first. In a previous project, our team had accumulated significant debt in our authentication layer. I proposed a phased refactoring plan that we executed over three sprints, which reduced related bug reports by over 60% and made the system significantly easier to extend.

6. What design patterns do you use most frequently?

Why interviewers ask this: They want to gauge your understanding of software design principles and your ability to apply them thoughtfully.

The patterns I use most depend on the context, but several come up regularly. I use the Factory pattern when I need to create objects without exposing instantiation logic — for example, creating different database connection types based on configuration. The Observer pattern is useful for event-driven architectures, and I've used it extensively in real-time notification systems.

For services that should have exactly one instance (like connection pools or configuration managers), I use the Singleton pattern carefully, being mindful of testability concerns.

I'm also a proponent of the Strategy pattern for swappable algorithms — it keeps code clean and makes behavior easy to modify without touching existing logic.

That said, I try not to apply patterns dogmatically. The goal is readable, maintainable code, not pattern gymnastics.

7. How do you handle disagreements about technical approaches with teammates?

Why interviewers ask this: They're evaluating your collaboration skills and ability to navigate conflict constructively.

When I disagree with a colleague on a technical approach, I start by making sure I fully understand their perspective. I ask questions and listen carefully — often there's context or a constraint I haven't considered. Then I clearly articulate my reasoning, focusing on trade-offs rather than opinions.

In one memorable case, a colleague and I disagreed about whether to use a microservice or monolithic architecture for a new service. Rather than debating abstractly, we each spent a day prototyping our approach with realistic constraints. When we compared results, we found that a modular monolith with clear domain boundaries gave us the simplicity benefits of a monolith with the organizational benefits of microservices. The hybrid approach was better than either original proposal.

8. Describe a time you optimized code performance.

Why interviewers ask this: They want to see your ability to identify bottlenecks and apply optimization techniques effectively.

I identified a data processing script that was taking over 4 hours to complete its daily run. Using a profiling tool, I found that the primary bottleneck was individual database queries being executed inside a nested loop — the classic N+1 query problem. The script was making thousands of individual queries when a handful of batch queries would suffice.

I refactored the code to batch database reads, load the required data into memory-efficient data structures, and process it in a single pass. I also added database indexes on the columns used for filtering. The result was a 75% reduction in execution time — from over 4 hours to under 1 hour — with no change to the output. I added performance benchmarks to our test suite so we'd catch regressions early.

9. How do you approach system design?

Why interviewers ask this: They're assessing your ability to think at a higher level about architecture, scalability, and trade-offs.

I start every system design exercise by clarifying requirements and constraints. What are the functional requirements? What are the non-functional requirements (latency, throughput, availability)? What's the expected scale? Without this, you're designing in a vacuum.

From there, I sketch the high-level components and their interactions — typically starting with the user-facing API, the core business logic layer, and the data storage layer. I

identify the most critical path and design for that first. Then I address scalability concerns: where are the bottlenecks? Do we need caching, load balancing, or sharding? I always discuss trade-offs explicitly. For example, choosing between consistency and availability in a distributed system, or between a relational and NoSQL database based on access patterns. I find that articulating trade-offs clearly is often more valuable than arriving at a “perfect” design.

10. What motivates you as a Software Engineer?

Why interviewers ask this: They want to understand your intrinsic drive and whether you'll be engaged and fulfilled in the role.

I'm motivated by the combination of creative problem-solving and tangible impact. There's a unique satisfaction in taking a complex, ambiguous problem and building an elegant solution that real people use. I love the iterative nature of software — shipping something, getting feedback, and making it better.

I'm also energized by learning. The fact that this field constantly evolves means I'm never bored. Whether it's a new framework, a different architectural pattern, or a completely new domain, there's always something to explore. And increasingly, I find deep motivation in mentoring — helping a junior engineer have their “aha” moment is incredibly rewarding.

Behavioral Interview Questions

Behavioral questions assess how you've handled real situations in the past. Use the STAR method (Situation, Task, Action, Result) to structure your answers clearly.

11. Tell me about a time you had to meet a tight deadline on a project.

Why interviewers ask this: They want to evaluate your time management, prioritization, and performance under pressure.

Situation: Our team was tasked with delivering a major API overhaul two weeks ahead of schedule because a key client accelerated their integration timeline.

Task: I was responsible for redesigning and implementing three core API endpoints while maintaining backward compatibility with existing consumers.

Action: I immediately broke the work into smaller deliverables and identified which endpoints had the highest client impact. I focused on those first, writing comprehensive tests alongside the code to avoid rework. I communicated daily with the client's engineering team to validate assumptions early and avoid building the wrong thing. I also identified lower-priority tasks that could be deferred and discussed this with my manager.

Result: We delivered all three endpoints on time, with full backward compatibility. The client integration went smoothly with zero breaking changes, and the early communication prevented what could have been a costly misunderstanding about the response format.

12. Describe a situation where you mentored a junior engineer.

Why interviewers ask this: They're assessing your leadership ability and willingness to invest in team growth.

Situation: A new junior engineer joined our team and was struggling with our codebase, which was large and poorly documented in some areas.

Task: I volunteered to be their onboarding buddy to get them to independent productivity within their first month.

Action: I set up daily 30-minute pairing sessions where we worked through real tickets together. Rather than just showing them solutions, I walked through my thought process — how I read code, how I trace bugs, how I decide which approach to take. I also created a “codebase tour” document that mapped the key modules and their relationships, which benefited the whole team.

Result: Within three weeks, the junior engineer was completing tickets independently and participating constructively in code reviews. They later told me that the pairing sessions were the single most helpful part of their onboarding. The codebase tour document became a standard part of our onboarding process.

13. Tell me about a project that failed or didn’t go as planned.

Why interviewers ask this: They want to see how you handle setbacks, take accountability, and learn from mistakes.

Situation: I led the development of a caching layer intended to improve our application’s response time. We designed it based on assumptions about access patterns that turned out to be incorrect.

Task: After deploying to staging, we discovered that the cache hit rate was only about 15% — far below the 70%+ we needed to justify the added complexity.

Action: Rather than pushing forward, I called a team meeting to reassess. We analyzed actual production access patterns and realized our assumptions were wrong — users accessed a much wider variety of data than we expected. I proposed a revised approach using a tiered caching strategy with different TTLs based on data volatility. I also advocated for adding access pattern monitoring before implementing, so we’d design based on data rather than assumptions.

Result: The revised caching strategy achieved an 80% hit rate and reduced average response times by 45%. More importantly, I learned to validate assumptions with real data before committing to an architectural approach. I now build in lightweight monitoring as a first step for any performance optimization work.

Technical Interview Questions

14. Explain the difference between a stack and a queue. When would you use each?

Why interviewers ask this: They’re testing your understanding of fundamental data structures.

A stack is a Last-In-First-Out (LIFO) data structure — the most recently added element is the first to be removed. Think of a stack of plates. Common operations are push (add to top) and pop (remove from top), both $O(1)$.

A queue is a First-In-First-Out (FIFO) data structure — elements are processed in the order they were added. Think of a line at a store. Common operations are enqueue (add to back) and dequeue (remove from front), both $O(1)$.

I'd use a stack for scenarios like undo/redo functionality, expression parsing, or depth-first search traversal. I'd use a queue for task scheduling, breadth-first search, or any processing pipeline where order matters — like a message queue in a distributed system.

15. How would you design a URL shortener?

Why interviewers ask this: They want to evaluate your system design skills, including scalability and trade-off analysis.

I'd start by clarifying the requirements. For a URL shortener, the core operations are: given a long URL, generate a short URL; given a short URL, redirect to the original. Key non-functional requirements include low latency for redirects, high availability, and the ability to handle billions of URLs.

Storage: I'd use a key-value store (like DynamoDB or Redis) where the key is the short code and the value is the original URL. For persistence, I'd back this with a relational database.

Short code generation: I'd use a base-62 encoding of an auto-incrementing ID or a hash-based approach. Base-62 (a-z, A-Z, 0-9) with 7 characters gives us ~3.5 trillion unique codes.

Read path: Short URL → lookup in cache (Redis) → if miss, lookup in database → redirect (301 or 302). Caching the most popular URLs handles the read-heavy workload.

Write path: Receive long URL → generate short code → store mapping → return short URL. I'd use a load balancer in front of multiple application servers.

Scalability: The read-to-write ratio is very high (maybe 100:1), so I'd optimize for reads with aggressive caching and database read replicas. For write scaling, I could partition the ID generation space across multiple servers.

16. What is the difference between SQL and NoSQL databases? How do you choose between them?

Why interviewers ask this: They're evaluating your understanding of data modeling and your ability to make pragmatic technology choices.

SQL databases (PostgreSQL, MySQL) are relational — data is stored in tables with defined schemas, and relationships are enforced through foreign keys. They excel at complex queries, transactions (ACID compliance), and data integrity. They're the right choice when your data has clear relationships, you need strong consistency, or you're running complex analytical queries.

NoSQL databases (MongoDB, DynamoDB, Cassandra) offer flexible schemas and are designed for specific access patterns. Document stores are great for hierarchical data, key-value stores for simple lookups at scale, and column-family stores for time-series or

wide-column data. They typically offer better horizontal scalability and performance for specific workloads.

My decision framework: If the data is highly relational with complex joins, I lean toward SQL. If I need massive scale with simple access patterns, or if the schema evolves frequently, I lean toward NoSQL. In practice, many systems use both — SQL for transactional data and NoSQL for caching, session storage, or analytics.

17. Explain how you would implement a CI/CD pipeline for a new project.

Why interviewers ask this: They want to assess your understanding of modern development practices and automation.

I'd design the pipeline in stages that provide progressively more confidence before deployment:

Stage 1 — Build: On every push to a feature branch, the pipeline compiles the code, resolves dependencies, and runs linting and static analysis. This catches syntax errors and style violations immediately.

Stage 2 — Test: Run the full unit test suite and integration tests. I'd parallelize tests where possible to keep feedback fast. Target: the entire build-and-test cycle completes in under 10 minutes.

Stage 3 — Security scan: Run dependency vulnerability scanning (e.g., Snyk or Dependabot) and static application security testing (SAST).

Stage 4 — Staging deployment: On merge to main, automatically deploy to a staging environment that mirrors production. Run end-to-end tests and smoke tests against staging.

Stage 5 — Production deployment: After staging validation, deploy to production using a blue-green or canary strategy to minimize risk. Monitor error rates and key metrics during rollout, with automatic rollback if thresholds are breached.

I'd use GitHub Actions or GitLab CI for the pipeline definition, Docker for consistent build environments, and infrastructure-as-code (Terraform) for managing the deployment targets.

18. What are microservices, and when would you choose them over a monolith?

Why interviewers ask this: They want to evaluate your architectural judgment and understanding of trade-offs.

Microservices architecture decomposes an application into small, independently deployable services, each responsible for a specific business capability. Each service has its own codebase, data store, and deployment pipeline. Services communicate via APIs (REST, gRPC) or message queues.

A monolith is a single deployable unit containing all the application's functionality. It's simpler to develop, test, and deploy — especially for smaller teams.

I'd choose microservices when: the team is large enough to own separate services (typically 50+ engineers), different parts of the system have different scaling needs, you

need independent deployment for faster release cycles, or the domain naturally decomposes into bounded contexts.

I'd stick with a monolith when: the team is small, the domain isn't well understood yet (premature decomposition is costly), or the operational overhead of managing distributed systems isn't justified. A common pragmatic approach is to start with a well-structured monolith and extract services as the need becomes clear.

- **<https://masterscareers.brown.edu/blog/2024/02/07/entry-level-software-engineer-in-interview-questions-with-answers>**

1. What is a whiteboard test?
 - *Answer/Explanation:* It's a live coding problem interview where you solve problems in real time (on a whiteboard or virtual environment). Interviewers evaluate how you approach the problem, not just whether the first solution is perfect.
 - *Tip:* Use a structured approach like UMPIRE (Understand, Match, Plan, Implement, Review, Evaluate) to walk through your thinking.
2. What programming languages do you use?
 - *Answer:* Know multiple languages — have one you master and 1–2 more you're strong in. Popular ones include JavaScript, HTML/CSS, Python, SQL, and TypeScript.
3. Describe the software development process.
 - *Answer:* The software development lifecycle includes:
 - i. Requirements gathering
 - ii. Design
 - iii. Development/Coding
 - iv. Testing
 - v. Deployment
 - vi. Maintenance
4. What different software development life cycle (SDLC) models are available?
 - *Answer:* SDLC models include Waterfall, Continuous Integration, Incremental Development, Rapid Application Development, and Agile.
5. What is agile software development?
 - *Answer:* Agile is a collaborative, incremental workflow that focuses on frequently delivering small pieces of working software and improving as you go.
6. Explain the difference between verification and validation.
 - *Answer:* Verification checks that a specific function works correctly; validation checks that the overall software meets user or client requirements.
7. Explain the difference between computer programs and computer software.
 - *Answer:* A computer program is a piece of code that performs a specific task. Computer software is a collection of programs, instructions, and documentation.
8. What is debugging?
 - *Answer:* Debugging is finding and fixing bugs in software. Typically, you first reproduce the issue, locate the cause in code, fix it, and then test again to confirm it's resolved.

9. What is software re-engineering?
 - *Answer:* Updating or improving existing software, such as adding features or refactoring code to make it better or more maintainable.
10. What is software scope?
 - *Answer:* Software scope defines the boundaries of a project — what it will do and what it will *not* do. This helps guide planning and resources.
11. What are CASE tools?
 - *Answer:* Computer-Aided Software Engineering tools that help automate parts of the software development lifecycle (like error checking in diagrams or auto-generating code).
12. How would you explain an API to a non-technical person?
 - *Answer:* An API (application programming interface) is like a *translator* that lets two programs communicate. For example, it connects a flight-search site to airline databases so users see flight options without manually checking each airline.
13. Why do you want to go into software engineering?
 - *Answer:* Explain your interest in the field — e.g., passion for technology, problem-solving, and building useful things. Tailor it to your personal experience.
14. Talk about a project you completed successfully.
 - *Answer:* Describe the project's purpose, your role, how you approached it (including teamwork and planning), and what the outcome was.
15. Tell me about a time you overcame a problem in software development.
 - *Answer:* Discuss a challenge (like a bug or deadline), steps you took to resolve it (testing, debugging, asking for help), and what you learned.
16. How do you keep your skills sharp and up to date?
 - *Answer:* Mention coding practice, personal or open-source projects, online courses, or learning new technologies to stay current.
17. Why do you want to work at this company?
 - *Answer:* Show that you researched the company's mission, values, projects, or products and explain why your interests and skills align.

- <https://www.geeksforgeeks.org/hr/hr-interview-questions>

1. Tell me about yourself.

I am a software developer with experience in full-stack development using MongoDB, Express.js, React, and Node.js. My projects range from small websites to large applications. What's especially important for me is solving problems in a manner that is useful for people in the process. I have graduated with a higher education in Computer Science. Throughout the university period, I tried to pay attention to both strongly developing my technical skills and learning how to work with people effectively while working in teams.

Afterwards, I graduated and joined [Company Name]. Working in a team, we developed [specific project or product] while working for the companies mentioned, which helped

me get stronger problem-solving abilities and be attentive to detail. I love challenges and equally relish the continuous learning of new technologies, which makes this opportunity quite exciting to me; I can only imagine the work your company is doing in relation to the industry or field, which is really innovative, and I'd love to chip in my skills and experience to help the team achieve its goals.

2. What are your strengths?

Problem-solving is one of my key strong sides: I like being pushed by complex challenges, breaking them down into manageable components. For example, during my last project, I had to optimize the web application that faced slowness while loading. I have identified a number of bottlenecks in the code and implementation of different caching strategies, which allowed improvement of performance by a factor of 30%. Another strong quality of mine is that I am very communicative. The reason I say this is because of how comfortable I feel working with cross-functional teams, be it discussing technical issues with developers or explaining project requirements to non-technical stakeholders. It is this bridging of gaps between different functional groups that helps me work efficiently across diverse environments. I am also quite conscientious in my work and always ready for further skill development through online courses or feedback from the team.

3. What are your weaknesses?

One area I have been working on is my tendency to take on too much at once. I'm very dedicated to delivering high-quality results, so I sometimes overcommit myself to ensure that I'm helping with a number of tasks or projects. However, I've realized this might lead to burnout, or at least a drop in efficiency. In fact, for such a case, I try to work on tools and techniques for good time management and prioritization, like the Eisenhower Matrix, so that I focus on activities that are urgent and important, while other less critical activities would have me promptly delegate or postpone them. In this way, I could sustain my productivity at work without any compromise on quality.

4. Why do you want to work here?

I've been following [company name] for some time and have been really impressed with your commitment to innovation and the quality of the products/services you offer. I'm particularly excited about the work you're doing in [specific industry or project], as it aligns closely with both my interests and my professional background. Projects that make a difference interest me, and I feel that my developed skills, particularly in [specific skill or technology], could serve as a great fit for your team. I also prefer a team-oriented environment, and your company culture, which emphasizes [MENTION COMPANY VALUES OR CULTURE, e.g., creativity, teamwork, or growth]-is just the kind of setup where I can contribute actively and grow professionally.

5. Where do you see yourself in five years?

In five years, I see myself having grown technically and professionally. I would like, probably, to reach deep expertise in full-stack development, particularly in such areas as cloud computing or AI, because I'm really eager to explore those fields further. I also want to perform a leadership activity, probably leading a team of developers or taking some project management role. At the end, I am going to look forward to continuing my

education, making contributions to projects of innovation, and carrying out such tasks in a company like yours that embraces personal growth and career advancement.

6. How do you handle stress and pressure?

I handle stress and pressure by maintaining organization and focus on the task at hand. In times of busyness, I make sure to break larger tasks down into smaller steps that are more digestible and prioritize those with regard to urgency and importance. I am a strong believer in time management, which is why I use facilitating tools such as to-do lists and project management tools to keep track of deadlines and make steady progress toward their completion.

It is also good for me if, when something overwhelms me, I take short breaks. This helps me always refresh and keep my perspective clear. If it gets too high, I don't hesitate to communicate with my team or manager, discussing workloads or finding solutions together. Overall, I consider stress an opportunity to flex tenacity and be able to stay focused on achieving the pre-set goals

7. Do you prefer working alone or in a team?

I am comfortable with both. Working alone allows me to control my activities, be focused, and concentrate on something without distraction. It gives me self-sufficiency and ensures that a project can be maintained right from its beginning to the end. Also, I like working in a team as it creates new ideas and views through collaboration.

I contribute to a team by being a good listener and offering my opinion in cases where it would be appropriate. I just love idea sharing; for me, teamwork really improves the quality of the work done. Of course, I am flexible and adapt where needed, to what the particular task or project may demand, be it working alone or as part of a team.

8. What motivates you?

I am motivated by the opportunity to solve problems and contribute to something meaningful. It could be in the form of developing any new feature, enhancing the working of any system, or learning a new technology. The challenging environment allows me to grow. My motivation also comes from the success of the teams that I work with. I like to contribute to projects that help the team or company reach goals, and it feels great to see positive results from our efforts.

For instance, what motivates me on a personal level is continuous learning. I look out for different skills and technologies that help me stay updated in the field, and I just love this feeling of progress when learning or mastering something new. It is that combination of problem-solving, personal growth, and teamwork that keeps me going.

9. How do you prioritize your tasks?

I tend to plan by prioritizing tasks based on their urgency and importance. I mean, typically it goes something like this: finding the deadline, breaking down projects into actionable tasks, and then categorizing those into must-haves, should-haves, could-haves, and won't-haves. For example, if a task is urgent and high-impact, it takes precedence.

I also put into consideration how long each task would take to carry out and the resources involved in doing so. Knowing that a particular task could take longer, I try to create sufficient time for it in order not to rush through, making sure it is well done. Often, I revisit my to-do list to reassess priorities, especially if new tasks or urgent requests float

in. I use the Eisenhower Matrix and some other time management techniques in conjunction with task management tools to help me stay on top of everything.

10. How do you handle failure?

I consider failure an opportunity for growth and learning. In the event of failure in my operations, I usually step back to see precisely what happened and what I could learn from that experience. I will ask myself a lot of questions: Did I mismanage my time? Was communication poor? Did I miss a key detail? Where there is an ability to perceive what exactly went wrong, it is really helpful to avoid making similar mistakes in the future.

I also look to ask for the feedback of my colleagues or superiors so that I can see it from a different perspective. Failure for me is never something to be afraid of; instead, it means an opportunity to revise my ways and come back more determined. I believe our response to failure is just as important as the success that follows.

HR Questions and Answers For Experienced

11. Can you describe a challenging work situation and how you overcame it?

In my previous job, we were tasked with delivering a large-scale project under a tight deadline. The challenge came when a key team member left midway through the project, which put additional pressure on the rest of the team. The project was already behind schedule, and we were at risk of missing the deadline.

To overcome this, I took the initiative to reallocate tasks within the team based on individual strengths. I also volunteered to take on extra responsibilities to fill the gap left by the departing team member. We worked closely together to break the remaining work into smaller, manageable chunks and prioritised the most critical elements of the project. I also kept open lines of communication with the client, managing their expectations and providing regular updates.

In the end, we were able to deliver the project on time. The experience taught me the importance of adaptability, clear communication, and teamwork under pressure.

12. What did you like most about your last job?

What I liked most about my previous job was the opportunity to work on varied and challenging projects. In each project, different problems were to be solved, keeping me out of being bored and continuously learning and growing. I also loved how the team collaborates. It was open, with ideas from everybody welcome, which makes working in such a positive and friendly atmosphere very motivating.

Besides, the company was keen on professional growth, which for them meant putting me through workshops and other training that would help me in building my technical skills. I appreciate and enjoy the attention to growth and collaboration.

13. What did you dislike about your last job?

In my past role, there were a lot of things that I enjoyed; however, limitations were minimal as far as career development opportunities were available. You see, since it is a small company, there were limited opportunities to enter leadership positions or climb up the career ladder in a very short period of time.

That said, I can appreciate it was due to the size and structure of the company, and I respected that. What it did do, though, is make me realize that I do enjoy an environment where there's room for professional growth and development-and that's actually something I am really looking for in my next role.

14. Why did you leave your last job?

Basically, I left my previous job because I reached a level where professional growth and the possibility of working on more projects were limited. Since I had experienced such great things and learned much from that place, I was looking for another challenge where I could put all my skills to work in a more dynamic setting, especially with opportunities for growth and career advancement.

I'm excited about this opportunity because it offers a chance to take on new responsibilities, work on more complex projects, and continue to develop both technically and professionally. I believe this role is a better fit for my long-term career goals

15. Describe your ideal work environment.

My perfect work environment would be collaborative, innovative, and requires continuous learning. I like to work in teams where open communication is established, and everyone can share their ideas with ease. I will also value a company culture supportive of creativity and problem-solving, in which individuals have sources to explore new solutions and take initiative. At the same time, I cherish the importance of balance; thus, I work best in flexible conditions that can offer both time for individual concentration and teamwork. Further, I do believe in maintaining a good work-life balance to sustain productivity and motivation in the long run. All in all, I feel great in an environment where professional growth can be attained, along with contribution to the team's success and support from leadership and colleagues in general.

16. What do you know about our company?

I have studied the company in great detail and find your reputation in the [industry/field] very impressive. Your commitment to [specific initiative, product, or service] really stands out to me because I've followed how you have continued to innovate in [mention a recent project, service, or market expansion]. Furthermore, I noticed that you care about mentioning [one of your company values: for example, sustainability, teamwork, or innovation], which is a key point of my professional values. I am looking forward to contributing to your further success, particularly regarding the work you are doing in [specific area related to the role].

17. How can you contribute to our company?

I firmly believe that I can contribute to your organization through a combination of technical expertise, problem-solving skills, and collaboration. My background in software development/project management experience will let me supportively contribute from the very first moment to your team through knowledge of relevant technologies/techniques, processes, or tools. I have always focused on the timely completion of high-quality work, so I am convinced that my ability to perform at a high level will keep me capable of supporting your group in its effort to.

I'm also dedicated to lifelong learning and constantly seeking ways to improve, which in turn enables me to introduce new processes or technologies that could promote productivity or efficiency.

18. What sets you apart from other candidates?

What makes me different is my unique combination of skills and experience. I bring a strong technical foundation in skills, but also a proven ability to work under conditions of

time pressure and velocity. Third and finally, I solve problems, always looking for creative, effective solutions.

What sets me apart is that I have great communication skills. I can bridge the gap between technical and non-technical people, making sure everyone is on the same page. That spirit of embracing new challenges in my life and continuous self-improvement helps me stay in front. I'm quite adaptable and open to feedback; this helps me grow both professionally and personally.

19. Are you willing to relocate/travel?

Yes, I am open to both relocation and travel if it's necessary for the role. I understand that some positions may require a bit of travel to attend client meetings, work with the team, and manage projects, which I am happy to accommodate. Similarly, relocation will be considered by me only if I can be closer to the main offices or work more easily because it is an in-person environment. I am open to all these adjustments that need to be made in order to support the needs of the company.

20. How do you align with our company's values?

From what I've learned about your company, there are many core values that we share in common, such as placing a strong emphasis on innovation, integrity, teamwork, and sustainability. For instance, I value continuous improvement and am always looking for ways to drive it; I also appreciate how well your company values collaboration, diversity, and work-life balance, which is important to me because I do my best work in environments where communication and teamwork are encouraged. I am confident that my professional approach and personal values are in line with the mission of your company, and I look forward to the opportunity to work in a team that shares the same values.

Behavioral HR Interview Questions and Answers

21. Can you give an example of a time you showed leadership?

I was entrusted with a project management assignment that involved the delivery of an essential feature for the client, who was in a rush. In the middle, some unexpected technical difficulties brought down the pace and frustrated the team.

I initiated this by stepping forward, facilitating a team meeting, and redistributing tasks within the team that were illustrated by individual strengths. In addition, I worked directly with the client to manage their expectations, keeping them informed of progress while assuring them we were doing everything we could to meet the deadline.

With clear direction provided, open communication, and a morale boost for the team, it was possible to overcome technical challenges on time delivery. This further developed in me leading through example, keeping myself focused without losing my cool under pressure, thus paving the way for others in collaboration to reach a common goal.

22. Describe a time when you resolved a conflict.

There has been some disagreement between two team members on one occasion as to how to approach the development of certain features. One wanted to implement new technology, exciting to him, although another felt that it would introduce too much risk given the project timeline. The conflict was causing delays and affecting team morale. I set up a meeting with both of them and listened to both their problems to understand the perspective of each. After weighing the pros and cons of each approach, I

highlighted a compromise whereby we could do a small prototype using new technology to test its viability without committing. In this way, we will have an idea if the new technology was worth the risk.

Sounding out this approach, both of our team members agreed upon the approach, and indeed, the prototype showed that the new technology would go well with the project. By listening actively, facilitating a solution, and encouraging collaboration, we resolved the conflict and moved forward productively

23. How do you handle constructive criticism?

I look at constructive criticism as a necessary ingredient required for growth, either personal or professional. Whenever feedback is available, I always ensure that I am open-minded, focusing on those particular areas that require improvement. For example, during one of the code reviews, there is an instance where a senior developer pointed out that a function I had written was a bit inefficient; the logic could be simplified. I initially thought the function was okay and would work, but I welcomed the feedback and took my time learning other efficient ways of coding.

I also sought clarification on how to improve and went through resources on optimisation techniques. Therefore, I rewrote the function and made the code efficient, hence fast in processing. I feel constructive criticism helps me know my blind spots and improves my performance. I embrace it as a learning opportunity.

24. What is the most significant achievement in your career?

The major achievement in my career was the management of a project that involved the migration of our entire infrastructure to a cloud-based system. This migration was critical, since our old system was inefficient and mostly down, thus interfering with serving our clients effectively.

I led the planning, execution, and coordination across diverse teams to make the transition as smooth as possible and cause minimal disruption to regular operations.

This involved detailed project management, effective communication with stakeholders, and overcoming several technical challenges regarding data integrity and system compatibility.

As a result of this migration, system performance increased by 40%, saving us a lot of downtime. This project not only demonstrated my leadership and problem-solving skills but also had direct, measurable results on company operations and customer satisfaction.

25. How do you stay updated with industry trends?

I keep myself abreast of what's happening in this industry through a variety of resources. I subscribe to a scant few newsletters-for example, mention a relevant newsletter, blog, or industry publication-that cover the latest trends, best practices, and new technologies. I also follow thought leaders and key industry players on business networking sites like LinkedIn and Twitter to keep my finger on the pulse. I also regularly join webinars, online courses, and conferences, both virtual and in-person, to add to my knowledge.

For example, recently I attended a webinar regarding, for instance, an event relevant for the industry or another, where I was updated about recent happenings concerning, for instance, specific technology or trend; some concepts I try to implement in my projects at the moment. By dedicating time to continuous learning and actively engaging with the

professional community, I ensure that I remain informed and adaptable to changes in the industry.

Panther's Q & A

- 1. As you're thinking about your next position, what are the top three things that are most important for you to find?**
- 2. If you could have changed anything about that position what would it have been?**
- 3. What did you like best about your position at Deepsync?**
- 4. Would you be able to go in through if it was necessary, like if there was a meeting that was taking place?**

If it's actually super necessary, I think that I should be able to go. There should be possible.

- 5. Have you faced any challenges working from home that differ from an in-office environment?**

Top's Q&A (Final round interview questions)

- 1. What does your ideal day at work look like? And what does your least ideal day at work look like?**
- 2. When priorities change suddenly, how do you handle that situation and stay effective?**
- 3. You have around 6 years of experience. Across different companies and teams, what team size do you work best in, and how do you collaborate with teams?**
- 4. If you are on an on-call rotation and a support or internal customer issue comes in, what is your first step and how do you approach it?**
- 5. How do you use AI in your daily work or personal workflow? How do you integrate it into your day-to-day process, and what things do you watch out for when using it?**
- 6. What is your approach to learning a new programming language or technology?**
- 7. What is your favorite programming language to work with, and why? What do you like most about it?**
- 8. Have you worked with microfrontend architecture before? What are your thoughts on it, and when do you think it is the right choice to use it?**
- 9. Have you ever been part of an on-call rotation or held a pager for high-visibility or customer-facing systems? If yes, how was your experience and how did you handle it?**